

EXECUTIVE SUMMARY

Customer Segmentation & Churn Pattern Analytics

PROJECT OVERVIEW

Customer churn represents one of the most critical challenges in modern business. Research consistently shows that acquiring a new customer costs five to seven times more than retaining an existing one. This project addresses this challenge through a comprehensive data analytics framework that combines Exploratory Data Analysis (EDA), customer segmentation, and machine learning-based churn prediction to deliver actionable intelligence for retention strategies.

The study leverages customer demographic, service, and account data to build a predictive pipeline culminating in an interactive Streamlit dashboard, enabling business stakeholders to monitor churn risk in real time.

METHODOLOGY

The project followed a structured, end-to-end analytics workflow:

- Data Collection & Preprocessing — cleaning, encoding, and handling missing values
- Exploratory Data Analysis (EDA) — statistical profiling and visual pattern discovery
- Feature Engineering — transformation and selection of predictive variables
- Model Development — Logistic Regression and Random Forest Classifier
- Model Evaluation — accuracy, precision, recall, and F1-score benchmarking
- Dashboard Deployment — interactive Streamlit application for business users

The Random Forest Classifier was selected as the final production model due to its superior predictive accuracy, robustness to outliers, and ability to rank feature importance.

KEY FINDINGS

Analysis of customer behaviour revealed four dominant churn drivers:

Contract Type
Top Churn Driver

Tenure (< 12 month)
High Risk Segment

Monthly Charges
Positive Correlation

Electronic Pay
Higher Churn Rate

Additional findings indicate that electronic payment methods correlate with elevated churn, and that the combination of contract type and tenure duration together account for the greatest variance in churn outcomes.

STRATEGIC RECOMMENDATIONS

Based on model outputs and segmentation insights, the following six strategies are recommended to reduce churn and strengthen customer lifetime value:

Strategy	Description
Promote Long-Term Plans	Encourage customers to choose annual or multi-year contracts with incentives to reduce month-to-month churn.
Targeted Retention Campaigns	Identify high-risk segments using the Random Forest model and deploy personalized outreach before churn occurs.
Early Onboarding Support	Strengthen customer support in the first 6-12 months when churn risk is highest.
Loyalty Rewards Program	Introduce rewards and discounts for long-tenure customers to reinforce retention.
Pricing Optimization	Monitor high monthly charge customers and offer tiered pricing or custom bundles to reduce cost-driven churn.
Predictive Analytics Integration	Embed the churn prediction model into CRM workflows for continuous, proactive monitoring.

PROJECT DELIVERABLES

Analytical Deliverable	Technical Deliverable
Exploratory Data Analysis Report	Random Forest Model (.pkl)
Customer Segmentation Analysis	Interactive Streamlit Dashboard
Churn Prediction Model	GitHub Repository & Documentation
Research Paper	Full Source Code

CONCLUSION

This project demonstrates that machine learning and data analytics can effectively decode customer churn behaviour and guide proactive retention strategies. The Random Forest model provides reliable churn predictions, while the interactive dashboard translates analytical output into decisions business teams can act on immediately. By implementing the six recommended strategies, particularly early customer support and long-term contract incentives, organizations can meaningfully reduce churn, improve customer satisfaction, and build more durable long-term relationships.
